

UNIVERSITY OF TRANSPORT HO CHI MINH CITY

INFORMATION SECURITY

OVERVIEW OF INFORMATION SECURITY

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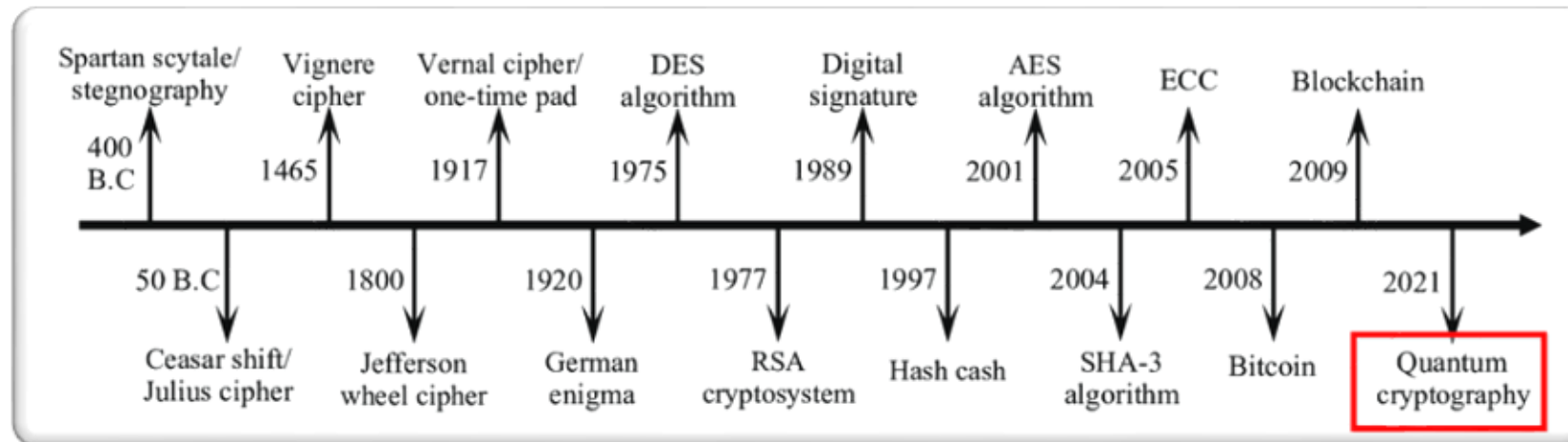
GENERAL KNOWLEDGE OF INFORMATION SECURITY



A BRIEF HISTORY OF CRYPTOGRAPHY

The history of cryptography is approximately 4,000 years old. The history of cryptography can be divided into four stages.

1. Naive cryptography
2. Formal cryptography
3. Scientific cryptography
4. Computer cryptography



GENERAL KNOWLEDGE OF INFORMATION SECURITY

THREATS IN INFORMATION SECURITY



First, we consider the threats to the system and classify them. Threats to the system can be classified into:

- **Unintentional threats:** when users restart the system in privileged mode, they can arbitrarily modify the system. However, after completing the work, they do not switch the system back to normal mode, unintentionally allowing malicious actors to exploit it.
- **Intentful threats:** such as intentionally accessing the system without authorization.
- **Passive threats:** are threats that have not yet or do not directly affect the system, such as eavesdropping on packets on the transmission line.
- **Active threats:** involve modifying information, changing the status, or the operation of the system.

GENERAL KNOWLEDGE OF INFORMATION SECURITY

POTENTIAL THREATS AND CONSEQUENCES

For every information system, potential threats and consequences are immense; they can originate from the following causes:

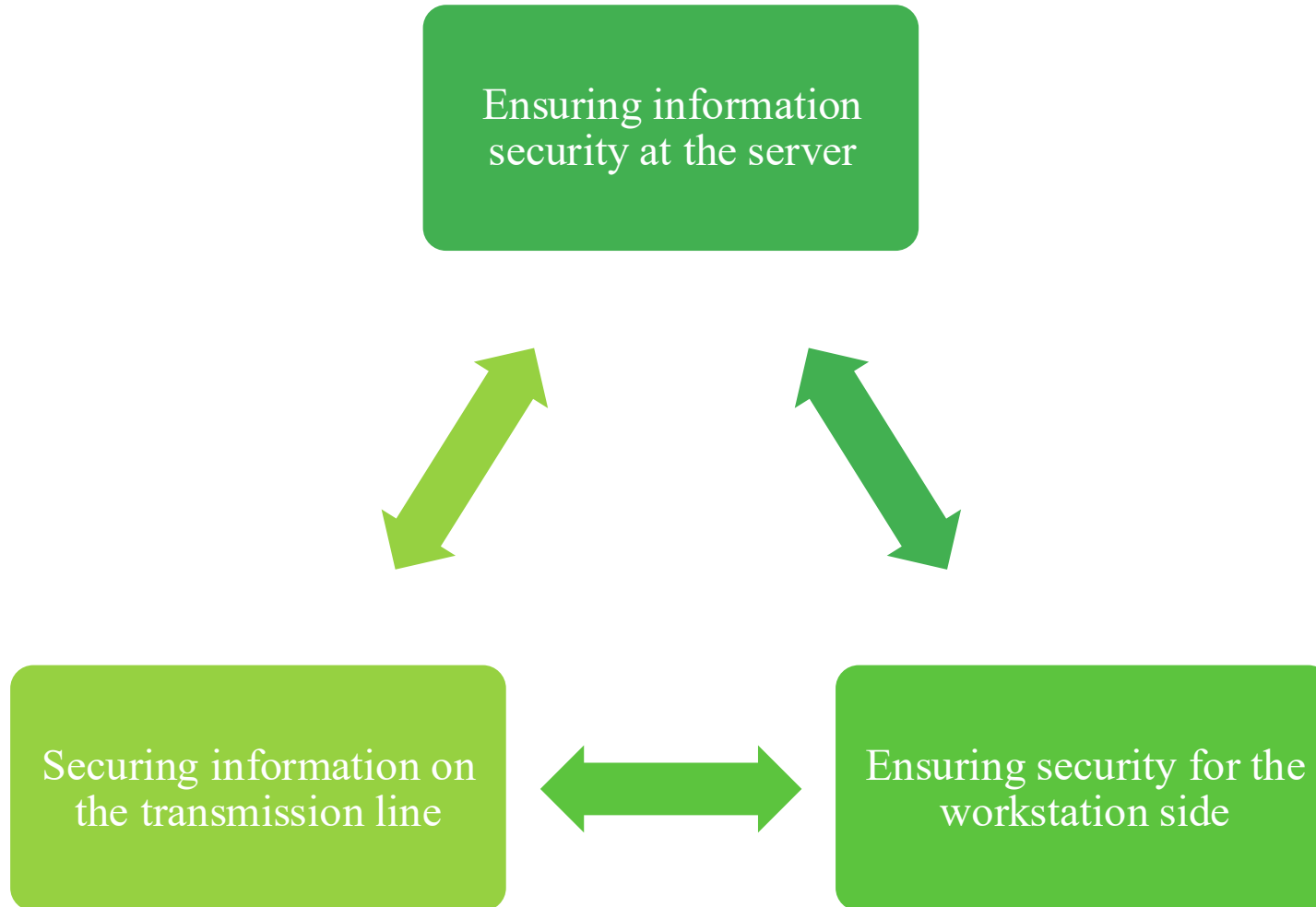
- **From the user side:** illegal intrusion, theft of valuable assets.
- **In information system architecture:** technical systems that are unstructured or not robust enough to protect information.
- **Within information security policies:** failure to comply with security standards, or failing to clearly define rights in system operations.
- **Information in computer systems** is also vulnerable to intrusion if there are no management, auditing, and system control tools.
- **Risks or vulnerabilities** reside within the hardware architecture of IT devices and in system or application software where manufacturers pre-install electronic "bugs" for intended purposes, known as "electronic bombs."
- **The most dangerous threat** to computer networks comes from hackers and criminal groups.



GENERAL KNOWLEDGE OF INFORMATION SECURITY



INFORMATION SECURITY TASKS



GENERAL KNOWLEDGE OF INFORMATION SECURITY

INFORMATION SECURITY SOLUTIONS

- Securing communication and network transmissions is far from simple. The requirements include confidentiality, authentication, non-repudiation, and integrity.
- **While developing security mechanisms and algorithms**, possible attacks must be considered. Sometimes, attackers view problems from a different perspective, allowing them to exploit system weaknesses.
- **Once security mechanisms are in place**, decisions must still be made regarding where to use them, on which protocols, on which devices, and through which services.
- **A typical security mechanism** involves multiple algorithms and protocols involving several parties ; for example, to perform encryption, both parties must share secret keys, as well as encryption and decryption algorithms.
- **Key distribution** between the sender and the receiver, along with the timeframes for transmission, must also be managed.



TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY



Information Protection - provides three main tasks:

- 1. Ensuring Confidentiality.**
- 2. Ensuring Information Integrity.**
- 3. Ensuring Information Availability.**

TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY

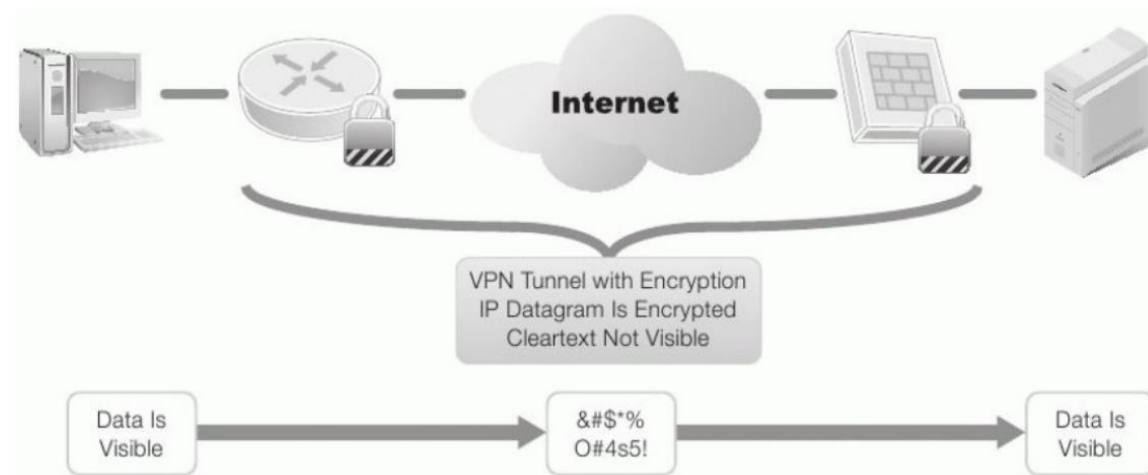


INFORMATION SECURITY REQUIREMENTS

Confidentiality: Only authorized users are allowed to access information.

Confidential information may include:

- Personal private data.
- Information belonging to the intellectual property of businesses or agencies/organizations.
- Information related to national security.



Confidentiality can be ensured through VPN encryption channels.

TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY

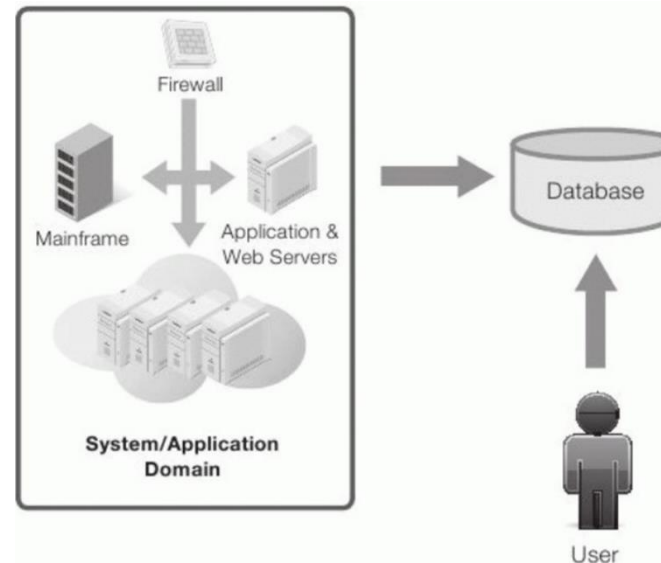
INFORMATION SECURITY REQUIREMENTS (CONTINUED)

Integrity: Information can only be modified by authorized users. Integrity relates to the validity and accuracy of data.

- In many organizations, information holds immense value, such as software copyrights, music copyrights, or patents and inventions.
- Any unauthorized changes can significantly impact the value of the information.

Data is considered integral if:

- The data has not been altered.
- The data is valid.
- The data is accurate.



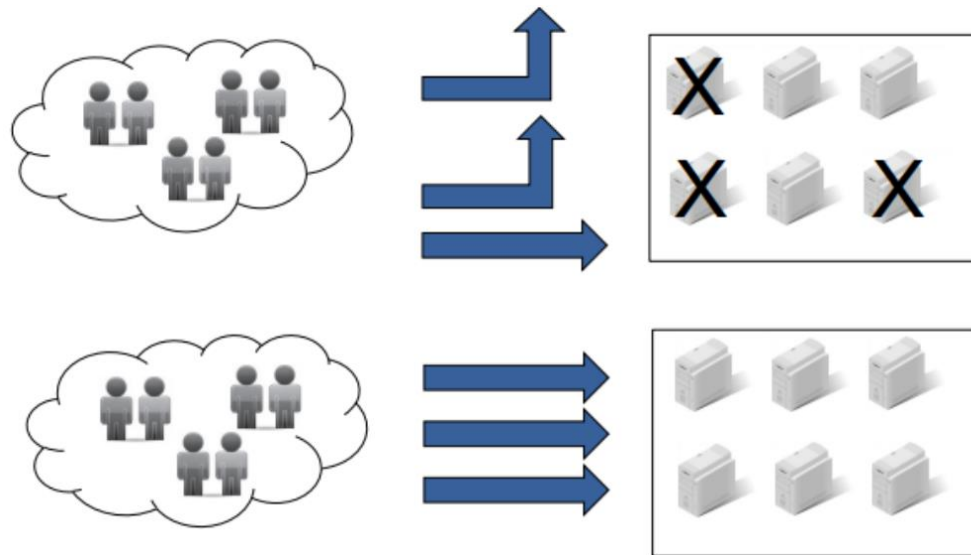
TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY

INFORMATION SECURITY REQUIREMENTS (CONTINUED)

Availability: Information can be accessed by legitimate users whenever they request it.

Availability can be measured by the following factors:

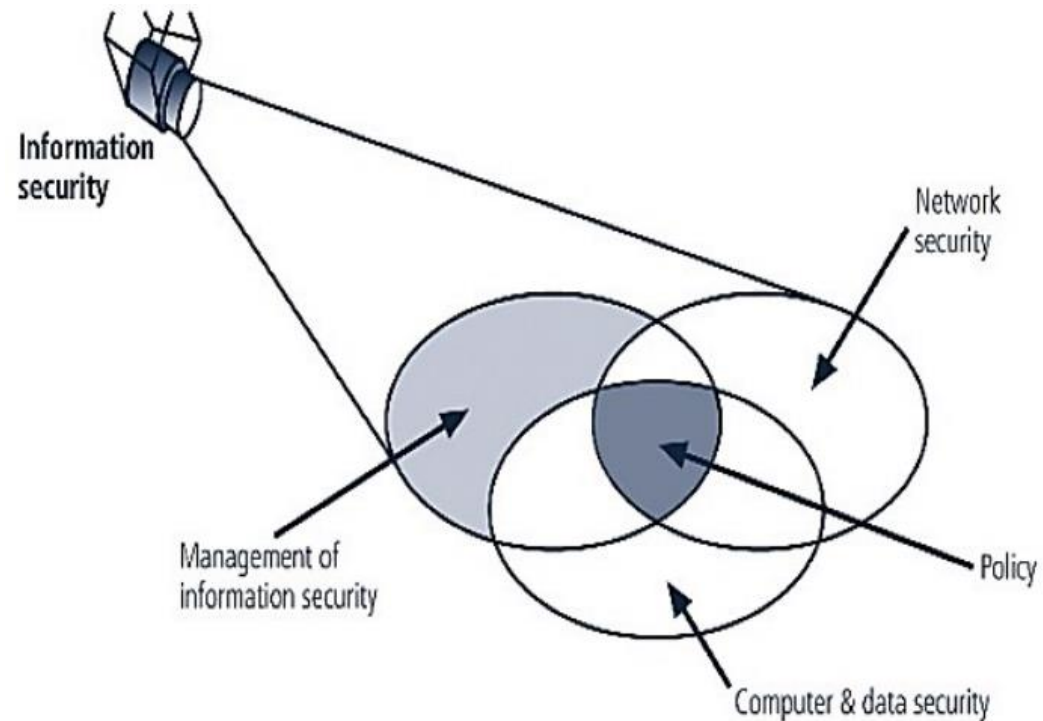
- Uptime (Service provision time)
- Downtime (Service interruption time).
- Service ratio: $A = (\text{Uptime}) / (\text{Uptime} + \text{Downtime})$.
- Mean Time Between Failures (MTBF).
- Mean Time To Repair (MTTR).
- Recovery time after an incident.



TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY

COMPONENTS OF INFORMATION SECURITY

- Computer and data security
- Network security
- Management of information security
- Information security policy



TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY



COMPONENTS OF INFORMATION SECURITY

COMPUTER AND DATA SECURITY:

- Ensuring the security of operating systems, applications, and services.
- Access control issues.
- Data encryption and security issues.
- Malware prevention and control issues.
- Data backup and redundancy, ensuring that data stored in the computer is not lost when incidents occur.



TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY



COMPONENTS OF INFORMATION SECURITY

NETWORK SECURITY:

- Firewalls and proxies for packet filtering and access control.
- Virtual Private Networks (VPN) and transmission security techniques such as SSL/TLS and PGP.
- Intrusion and attack detection and prevention systems and techniques;
- Network monitoring issues.

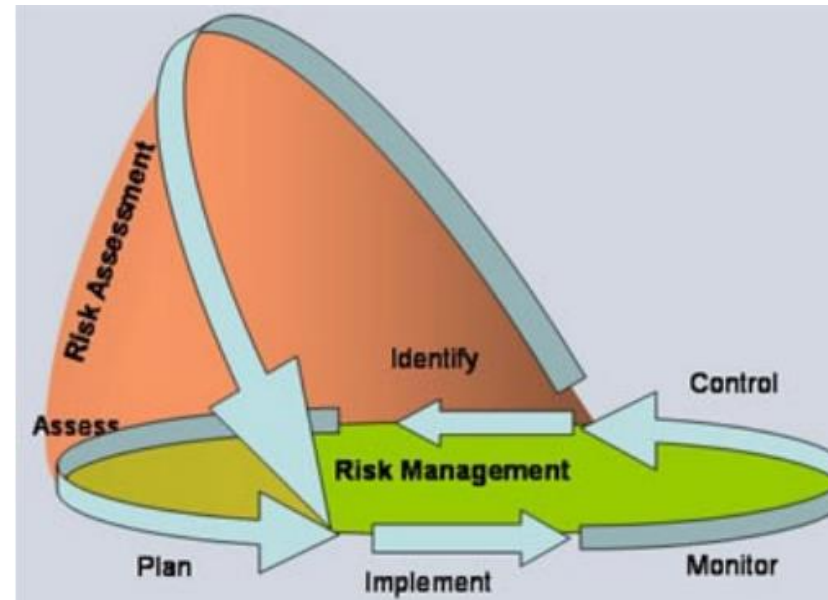


TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY

COMPONENTS OF INFORMATION SECURITY

MANAGEMENT OF INFORMATION SECURITY:

- Risk Management
 - Identification
 - Assessment
- Information Security Management Implementation
 - Plan
 - Do/Implement
 - Monitor
 - Control.



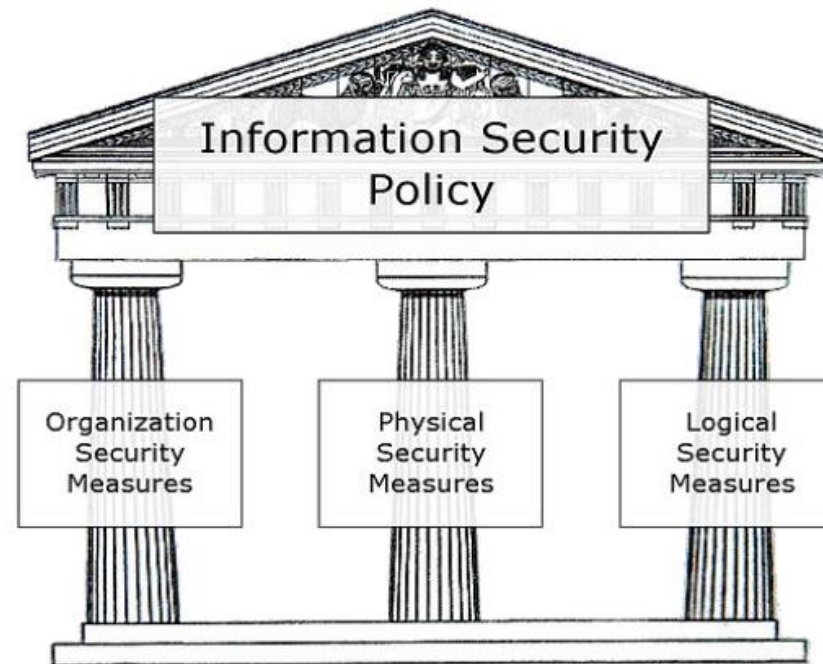
TECHNICAL CHARACTERISTICS OF INFORMATION SECURITY



COMPONENTS OF INFORMATION SECURITY

INFORMATION SECURITY POLICY:

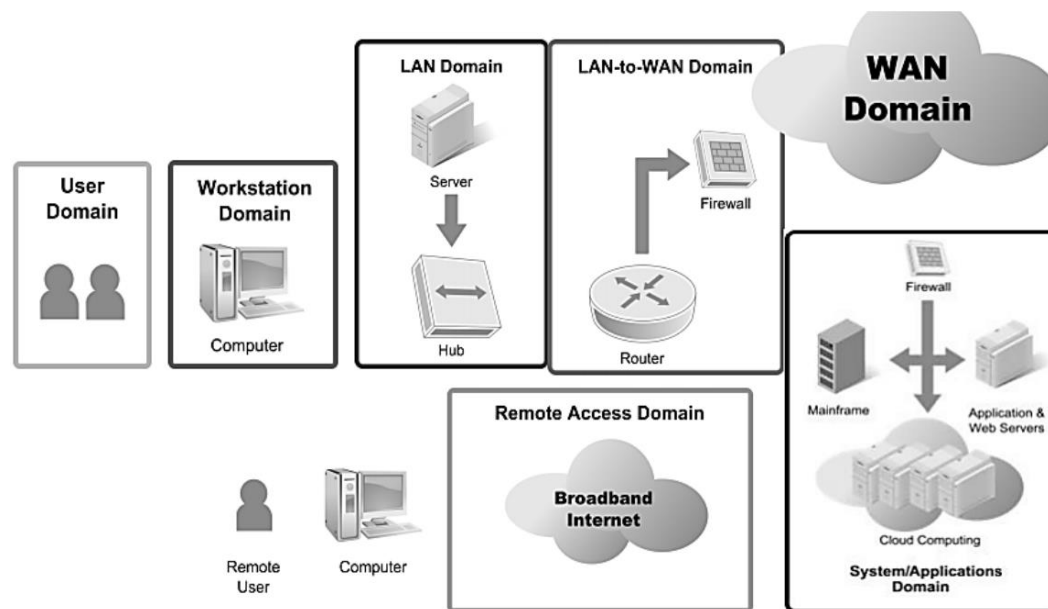
- Physical security policy.
- Organizational security policy.
- Logical security policy.



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

IT INFRASTRUCTURE DOMAINS:

- User domain
- Workstation domain
- LAN domain
- LAN-to-WAN domain
- WAN domain
- Remote Access domain
- Systems/Applications domain



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

THREATS TO THE USER DOMAIN:

- Lack of security awareness
- Disregard for security policies
- Security policy violations
- Inserting unauthorized removable media (portable HDD/USB) with personal files into the system
- Unauthorized downloading of images, music, and videos
- Sabotage of data, applications, and systems
- Malicious attacks from disgruntled employees
- Extortion or misappropriation of sensitive information by employees.



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

THREATS TO THE WORKSTATION DOMAIN:

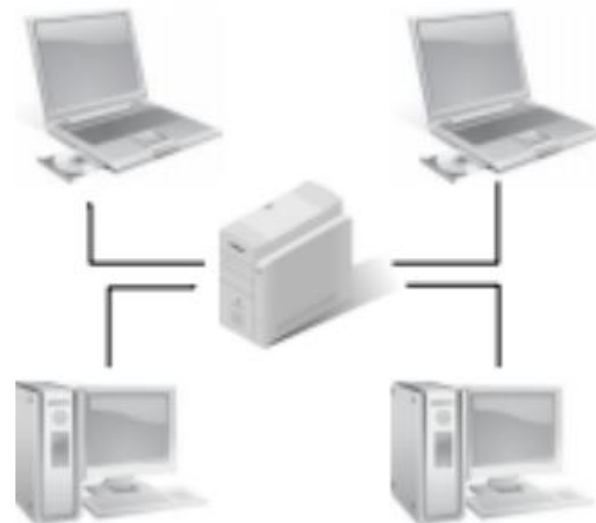
- Unauthorized access to workstations
- Unauthorized access to systems, applications, and data
- Security vulnerabilities in workstation operating systems
- Security vulnerabilities in workstation operating systems
- Threats from viruses, malware, and malicious software.
- Users inserting removable media (CD/DVD/USB) with personal files into the system
- Users downloading images, music, and videos



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

THREATS TO THE LAN DOMAIN:

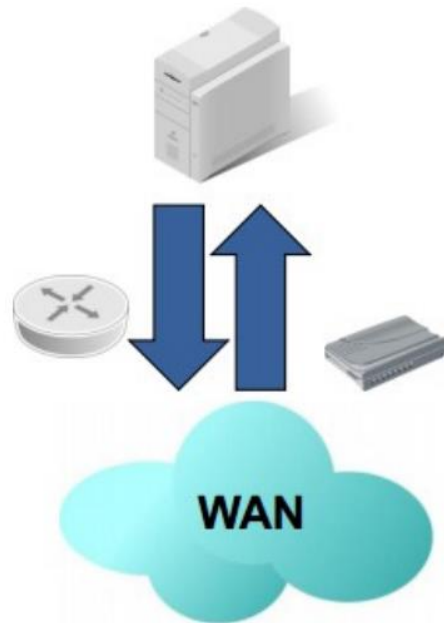
- Unauthorized physical access to the LAN
- Unauthorized access to systems, applications, and data
- Security vulnerabilities in server operating systems
- Security vulnerabilities in server application software
- Risks from rogue users in the WLAN
- Threats to data confidentiality in the WLAN
- Non-compliance with LAN server configuration standards and guidelines.



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

THREATS TO THE LAN-TO-WAN DOMAIN:

- Unauthorized probing and scanning of service ports
- Unauthorized access
- Security vulnerabilities in routers, firewalls, and other network devices
- Local users (within the LAN) downloading files of unknown content from untrusted sources.



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

THREATS TO THE WAN DOMAIN:

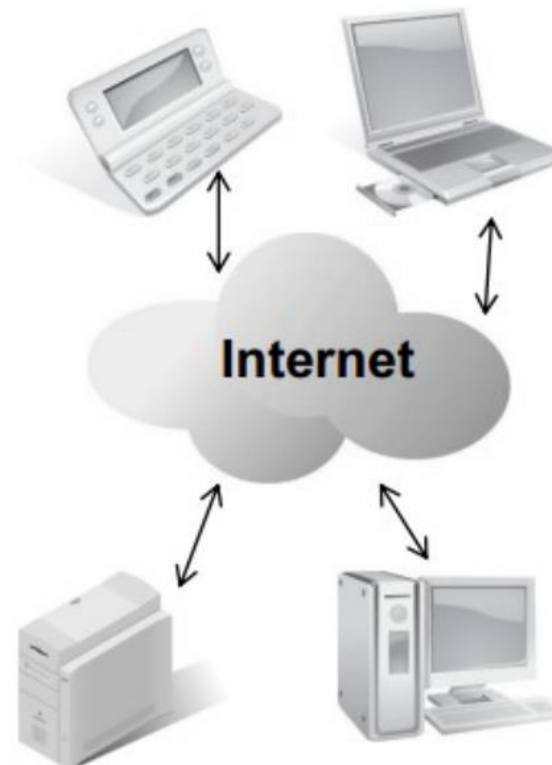
- Risks from data being accessible in an open and public environment
- Most data is transmitted in cleartext (plaintext)
- Vulnerability to eavesdropping
- Vulnerability to malicious attacks
- Vulnerability to Denial of Service (DoS) and Distributed Denial of Service (DDoS)
- Attackers can freely and easily send emails with attached viruses, worms, and other malware



THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS

THREATS TO THE REMOTE ACCESS DOMAIN:

- Brute force attacks on usernames and passwords
- Attacks on logon and access control systems
- Unauthorized access to IT systems, applications, and data
- Confidential information may be stolen remotely
- Data leakage due to violation of data classification standards.

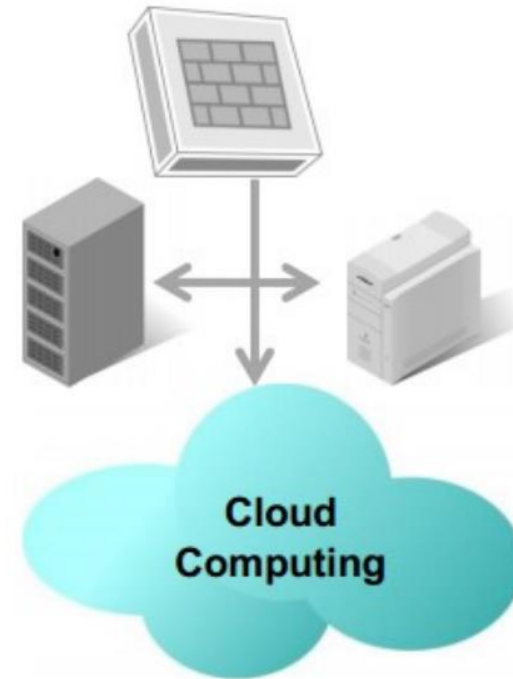


THREATS & RISKS IN IT INFRASTRUCTURE DOMAINS



THREATS TO SYSTEMS/APPLICATIONS ZONES:

- Unauthorized access to data centers, server rooms, or wiring closets
- Difficulties in managing servers that require high availability.
- Vulnerabilities in managing application software on host operating systems.
- Security issues within cloud computing virtual environments
- Issues of data corruption or data loss.



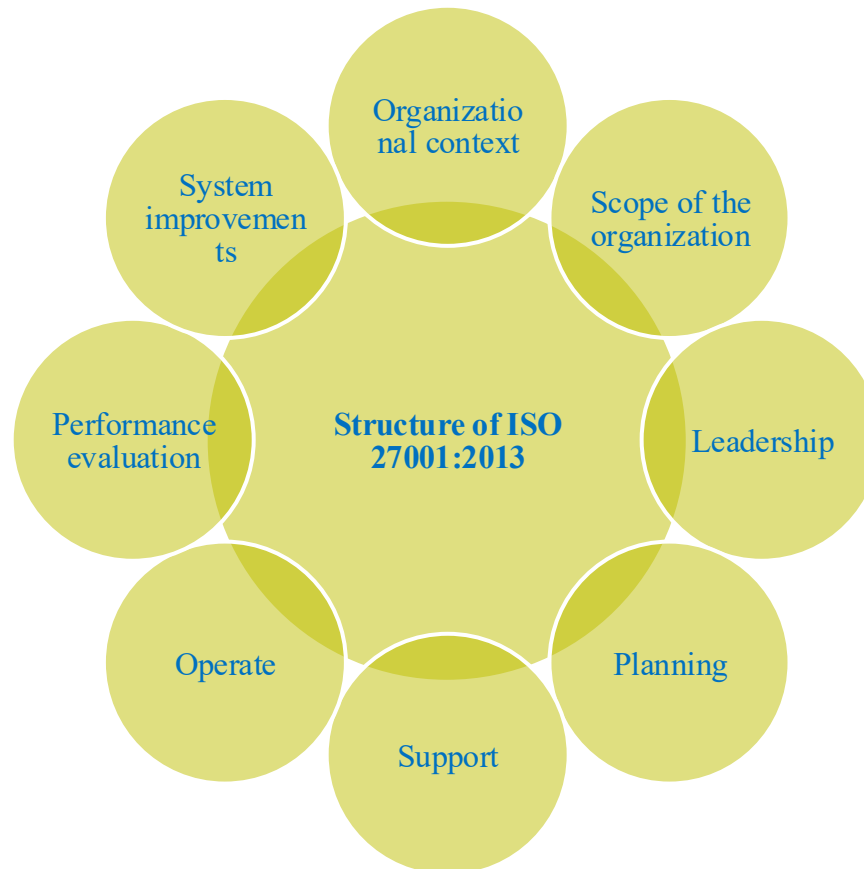
INFORMATION SECURITY STANDARDS



ISO = International Organization for Standardization

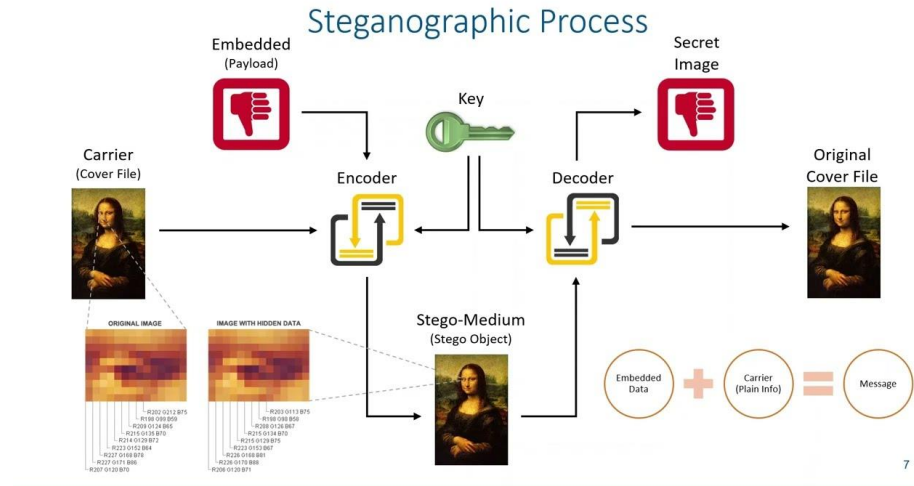
ISO 27001 is a world-renowned standard. In fact, it is the global standard for Information Security Management Systems (ISMS).

ISO 27001 ensures that a company or non-profit organization understands where its strengths and weaknesses lie.



INFORMATION SECURITY AND SAFETY TECHNIQUES

Steganography



7

Cryptography

Digital signature

Encryption & Decryption

